Merve Ileri-Tayar

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Academic Positions & Education

2026-present Postdoctoral Research Associate
Department of Psychology & Neuroscience
Duke University, NC, USA
Advisor: Tobias Egner

2021-2026 Ph.D., Psychological & Brain Sciences
Washington University in St. Louis, MO, USA
Advisor: Julie M. Bugg
Committee: Todd Braver, Wouter Kool, Richard Abrams, Jonathan Bogard

2021-2023 M.A., Psychological & Brain Sciences
Washington University in St. Louis, MO, USA
Advisor: Julie M. Bugg
Committee: Wouter Kool, Richard Abrams

2018-2021 M.Sc., Cognitive Psychology
Middle East Technical University, Ankara, Turkey
Advisors: Mine Misirlisoy, Nart Bedin Atalay
Committee: Asli Kilic Ozhan, Didem Kadihasanoglu

2014-2018 B.A., Psychology
TOBB University of Economics and Technology, Ankara, Turkey

Honors & Awards

2021-2026 McDonnell International Scholars Academy Fellowship
2026 Selected Graduate Student Speaker, 2026 Commencement Ceremony, Washington University in St. Louis
2026 McDonnell International Scholars Academy PhD-in-3D Competition (Runner-up)
2025 Dean's Award for Graduate Research Excellence (Winner), Washington University in St. Louis
2025 Departmental Dissertation Grant, Washington University in St. Louis
2024 Psychonomic Society Graduate Travel Award
2022 Psychonomic Society J. Frank Yates Student Conference Award
2021 Fulbright Ph.D. Grant (Awarded, Declined)

2020-2021 Middle East Technical University M.Sc. Thesis Project Grant
2018-2021 2210-A National Scholarship Program for M.Sc. Students, The Scientific and Technological Research Council of Turkey (TUBİTAK)
2014-2018 Full-Merit Scholarship, Department of Psychology, TOBB University of Economics and Technology, Turkey

Publications

* undergraduate mentee

Ileri-Tayar, M., Bugg, J. M., & Kool, W. (2026). Switching between cognitive control states? No, thank you. *Journal of Experimental Psychology: General, 155*(7), 1726-1747. https://doi.org/10.1037/xge0001940

Ileri-Tayar, M., Bugg, J. M., Dudey, T., & Braver, T. S. (2025). Proactive control declines while reactive control is preserved across the adult lifespan. *Journal of Experimental Psychology: General, 154*(11), 3029-3047. https://doi.org/10.1037/xge0001824

Nuño, C., **Ileri-Tayar, M.**, & Bugg, J. M. (2025). Attention on demand: Do people strategically heighten control when distraction is expected but rare? *Psychological Research*, *89*, 106. https://doi.org/10.1007/s00426-025-02131-2

Ileri-Tayar, M., Suh, J., Stern, A.*, Whitsitt, L.*, & Bugg, J. M. (2025). Does an external distractor interfere with the triggering of item-specific control? *Journal of Experimental Psychology: Human Perception and Performance, 51*(6), 808–825. https://doi.org/10.1037/xhp0001323

Ileri-Tayar, M., Colvett, J. S., Dey, A., & Bugg, J. M. (2025). Does “item-specific” cognitive control operate at the item level? *Journal of Experimental Psychology: Learning, Memory, and Cognition, 51*(8), 1238–1258. https://doi.org/10.1037/xlm0001432

Ileri-Tayar, M., Colvett, J. S., & Bugg, J. M. (2024). Between-task transfer of item-specific control is replicable and extends to novel conditions. *Journal of Experimental Psychology: Human Perception and Performance*, *50*(6), 535-553. https://doi.org/10.1037/xhp0001200

Ileri-Tayar, M., Moss, C.*, & Bugg, J. M. (2022). Transfer of learned cognitive control settings within and between tasks. *Neurobiology of Learning and Memory*, *196*, 107689. https://doi.org/10.1016/j.nlm.2022.107689

Suh, J., **Ileri-Tayar, M.**, & Bugg, J. M. (2022). When global and local information about attentional demands collide: evidence for global dominance. *Attention, Perception, & Psychophysics*, *84*(6), 1858-1873. https://doi.org/10.3758/s13414-022-02521-3

Manuscripts Submitted for Publication

Bugg, J. M., & **Ileri-Tayar, M.** (under review). When and how cognitive control is automatic. *Trends in Cognitive Sciences.*

Manuscripts in Preparation

Ileri-Tayar, M. & Egner, T. (in preparation). Item-level control in multitasking environments.

Ileri-Tayar, M. & Bugg, J. M. (in preparation). Learning and transfer of item-reward associations in conflict tasks.

Ileri-Tayar, M., Dey, A., Kool, W., & Bugg, J. M. (in preparation). Reward-induced shifts in cognitive control.

Keha, E., **Ileri-Tayar, M.**, Dreisbach, G., & Bugg, J. M. (in preparation). Age-related changes in task-switching: Distinct underlying mechanisms.

Ileri-Tayar, M. & Bugg, J. M. (in preparation). The puzzle of context-level reward learning.

Ileri-Tayar, M. & Bugg, J. M. (in preparation). Calibrating attention to graded distraction: The item-specific distraction intensity effect.

Invited Talks

Ileri-Tayar, M. & Bugg, J. M. (March, 2026). Learning and transfer of adaptive cognitive control. Invited talk at Rothermund Lab, Friedrich Schiller University Jena, Germany.

Bugg, J. M., & **Ileri-Tayar, M.** (July, 2025). The automaticity and flexibility of learning-guided item-specific control. Invited talk presented virtually at the Symposium on Conflict Adaptation across Single- and Multi-Tasking Paradigms, RWTH Aachen University, Aachen, Germany.

Ileri-Tayar, M. & Bugg, J. M. (May, 2024). Between-task transfer of learned control settings: How far can it go? Invited talk at Dreisbach Lab, University of Regensburg, Germany.

Ileri-Tayar, M. & Bugg, J. M. (December, 2023). Between-task transfer of learned control settings. Invited talk at Braem Lab, Ghent University, Belgium.

Talks at Conferences

Ileri-Tayar, M., & Bugg, J. M. (November, 2026). Calibrating attention to graded distraction: The item-specific distraction intensity effect. Talk to be presented at 67th Annual Meeting of the Psychonomic Society. San Diego, California.

Ileri-Tayar, M., Kool, W., & Bugg, J. M. (November, 2026). When reward meets conflict: How incentives reshape learned control. Talk to be presented at the Motivated Cognition Meeting. San Diego, California.

Ileri-Tayar, M., & Bugg, J. M. (March, 2026*).* Within- and between-task transfer of item-reward associations in conflict tasks. Talk presented at 68th TeaP (Conference of Experimental Psychologists). Tübingen, Germany.

Ileri-Tayar, M., Dey, A., Bugg, J. M., & Kool, W. (November, 2025). Target enhancement or distractor suppression? Modeling the source of reward-induced control enhancements. Talk presented at the Future of Cognitive Control Meeting. Denver, Colorado.

Ileri-Tayar, M., Bugg, J. M., & Kool, W. (September, 2025). Switching between cognitive control states? No, thank you. Talk presented at the 24th Biennial Conference of the European Society for Cognitive Psychology. Sheffield, United Kingdom.

Ileri-Tayar, M., Kool, W., & Bugg, J. M. (November, 2024). Reward induces shifts in learning-guided reactive control. Talk presented at the 65th Annual Meeting of the Psychonomic Society. New York City, New York.

Atalay, N. B., Bozkurt, O., & **Ileri-Tayar, M**. (October, 2022). Investigation of list-wide proportion congruence effect with response tracking methods. Talk presented at the 21st National Psychology Congress, Istanbul, Turkey.

Ileri, M., & Atalay, N. B. (November, 2018). Is asymmetrical list-shifting effect observed with item-specific shifting? Talk presented at the 20th National Psychology Congress, Ankara, Turkey.

Conference Poster Presentations

Ileri-Tayar, M., & Bugg, J. M. (November, 2025). When more information hurts: explicit reward information prevents transfer. Poster presented at the 66th Annual Meeting of the Psychonomic Society. Denver, Colorado.

Ileri-Tayar, M., Bugg, J. M., & Kool, W. (November, 2024). Switching between cognitive control states? No, thank you. Poster presented at the 65th Annual Meeting of the Psychonomic Society. New York City, New York.

Ileri-Tayar, M., Kool. W., & Bugg, J. M. (March, 2024). Does reward induce shifts in learning-guided reactive control? Poster presented at the 66th TeaP (Conference of Experimental Psychologists). Regensburg, Germany.

Ileri-Tayar, M., Bugg, J. M., & Braver, T. S. (November, 2023). A within-subjects comparison dissociating proactive and reactive control in younger, middle-aged, and older adults. Poster presented at the 64th Annual Meeting of the Psychonomic Society. San Francisco, California.

Ileri-Tayar, M., Moss, C., & Bugg, J. M. (November, 2022). Transfer of learned cognitive control settings within and between tasks. Poster presented at the 63rd Annual Meeting of the Psychonomic Society. Boston, Massachusetts.

Ic, Y., **Ileri-Tayar, M.**, & Atalay, N. B. (October, 2022). The influence of expectation on list-shifting: Initial findings. Poster presented at the 21st National Psychology Congress, Istanbul, Turkey.

Ileri, M., Misirlisoy, M., & Atalay, N. B. (May, 2020). Does response type change the Stroop effect in a mouse-tracking experiment? Poster presented virtually at the International Symposium on Brain and Cognitive Science.

Ileri, M., Misirlisoy, M., & Atalay, N. B. (September, 2019). Investigation of the Stroop effect with response tracking: Influence of response type. Poster presented at the 5th Experimental and Cognitive Psychology Symposium, Istanbul, Turkey.

Teaching Experience

2024 Instructor, Washington University in St. Louis, USA
Introduction to R (Summer course)

2023-2024 Assistant in Instruction, Washington University in St. Louis, USA
Graduate courses:
Quantitative Methods 1, Quantitative Methods 2

2019-2021 Teaching Assistant, Middle East Technical University, Turkey
Undergraduate courses:
Open Science and Replicability, Topics in Experimental Psychology: Brain and Behavior, Experimental Design & Analysis, Memory
Graduate courses:
Experimental Design & Analysis

Mentoring

Undergraduate research mentees, Washington University in St. Louis:

- Caroline Moss, Mind, Brain, and Behavior program. Coauthor, *Neurobiology of Learning and Memory* (2022); Psychonomics 2022 poster
- Amina Stern, Mind, Brain, and Behavior program. Coauthor, *JEP: HPP* (2025)
- Peter Ochalek, Mind, Brain, and Behavior program. Research assistant, *Psychological Research (2025)*
- Logan Whitsitt, Research assistant. Coauthor, *JEP: HPP* (2025)

Supervised seven additional undergraduate research assistants (2021-2026) in experiment data collection and analysis.

Professional Memberships

Member, Psychonomic Society
Associate Member, European Society for Cognitive Psychology
Women+ in Control

Reviewing

Ad-Hoc Reviewer (journals in alphabetical order):

Acta Psychologica
Cognition
Journal of Experimental Psychology: Human Perception and Performance
Journal of Experimental Psychology: Learning, Memory, and Cognition
Memory & Cognition
Psychological Research
Psychonomic Bulletin & Review

Service Activities

Faculty Search Committee Member, Psychological & Brain Sciences, Washington University in St. Louis, 2024-2025
Co-chair, McDonnell Academy Panel of Scholars (MAPS), 2024-2025
Committee Member, McDonnell Academy Panel of Scholars (MAPS), 2023-2026
Department Diversity Committee, Washington University in St. Louis, 2022-2023
Social Media Committee, Psychology Graduate Student Association, 2022-2023
Peer Mentoring Committee, Psychology Graduate Student Association, 2022-2023